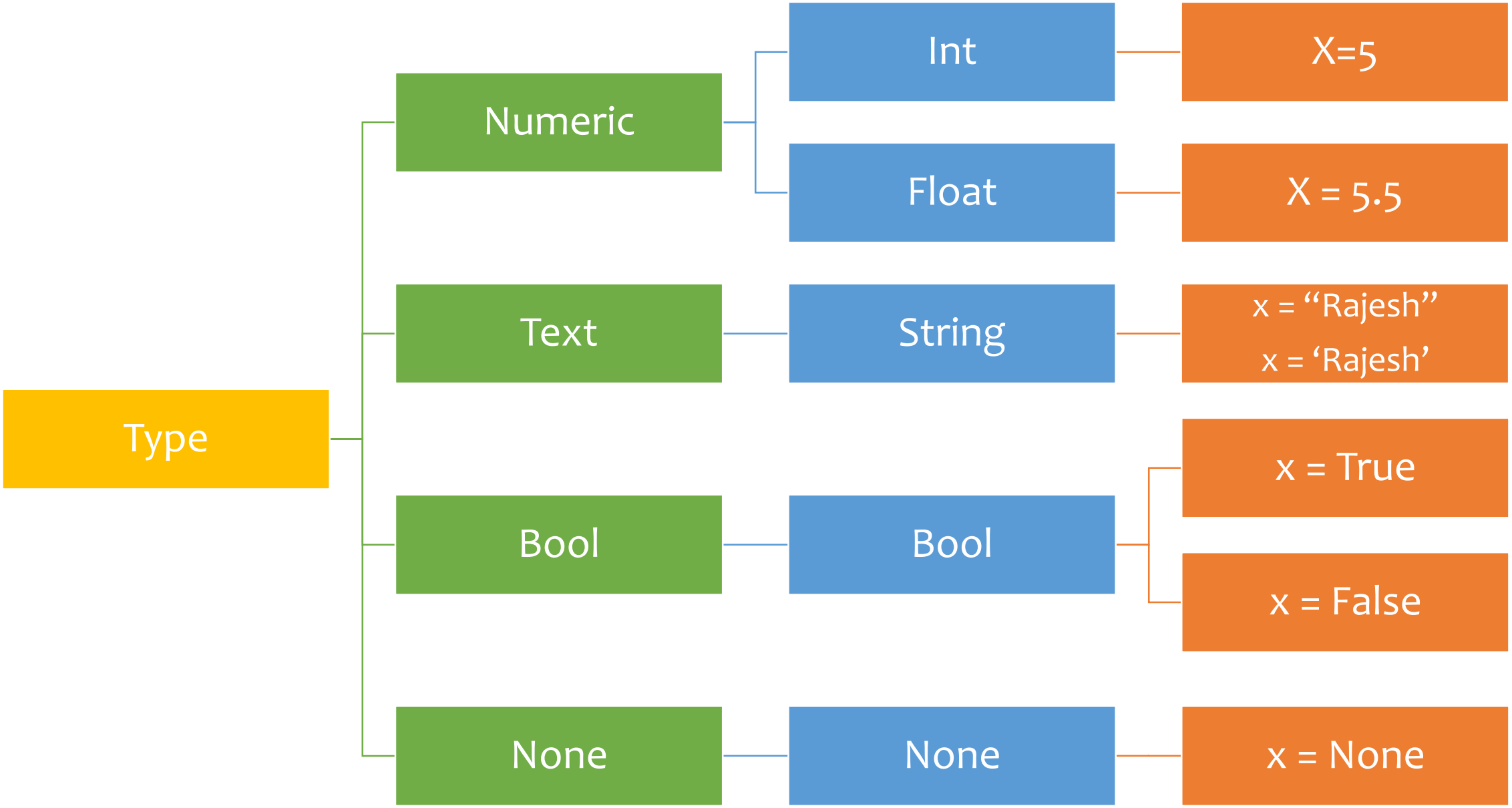
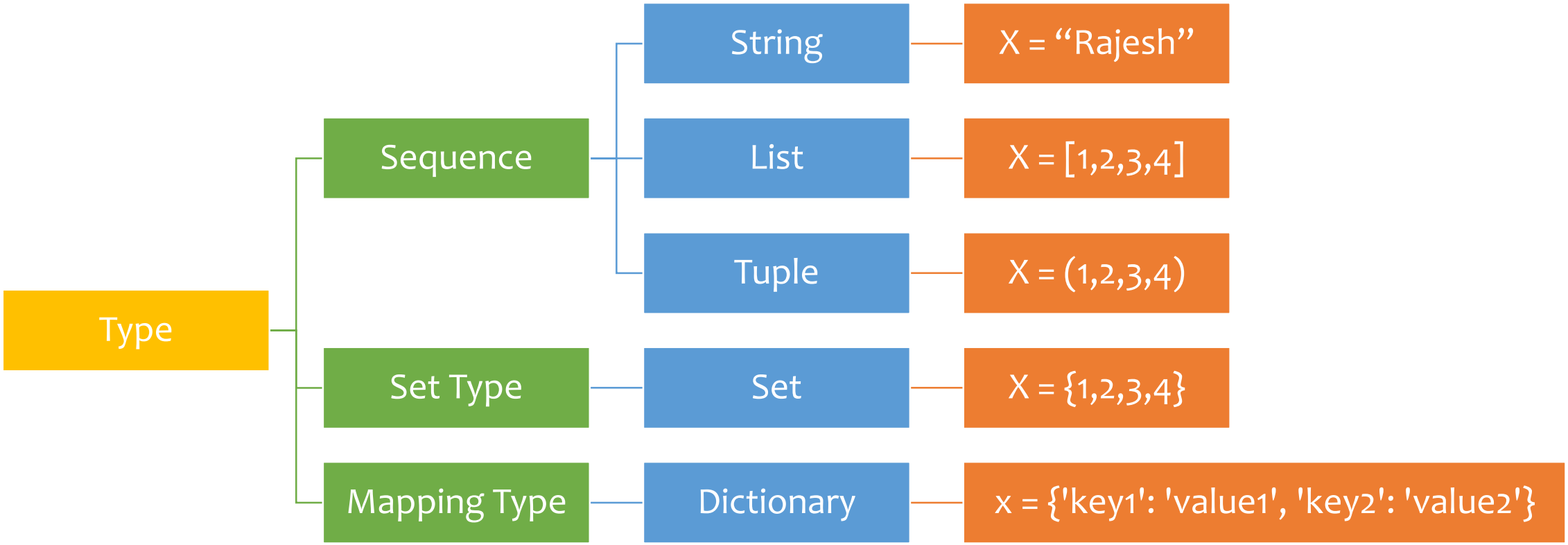
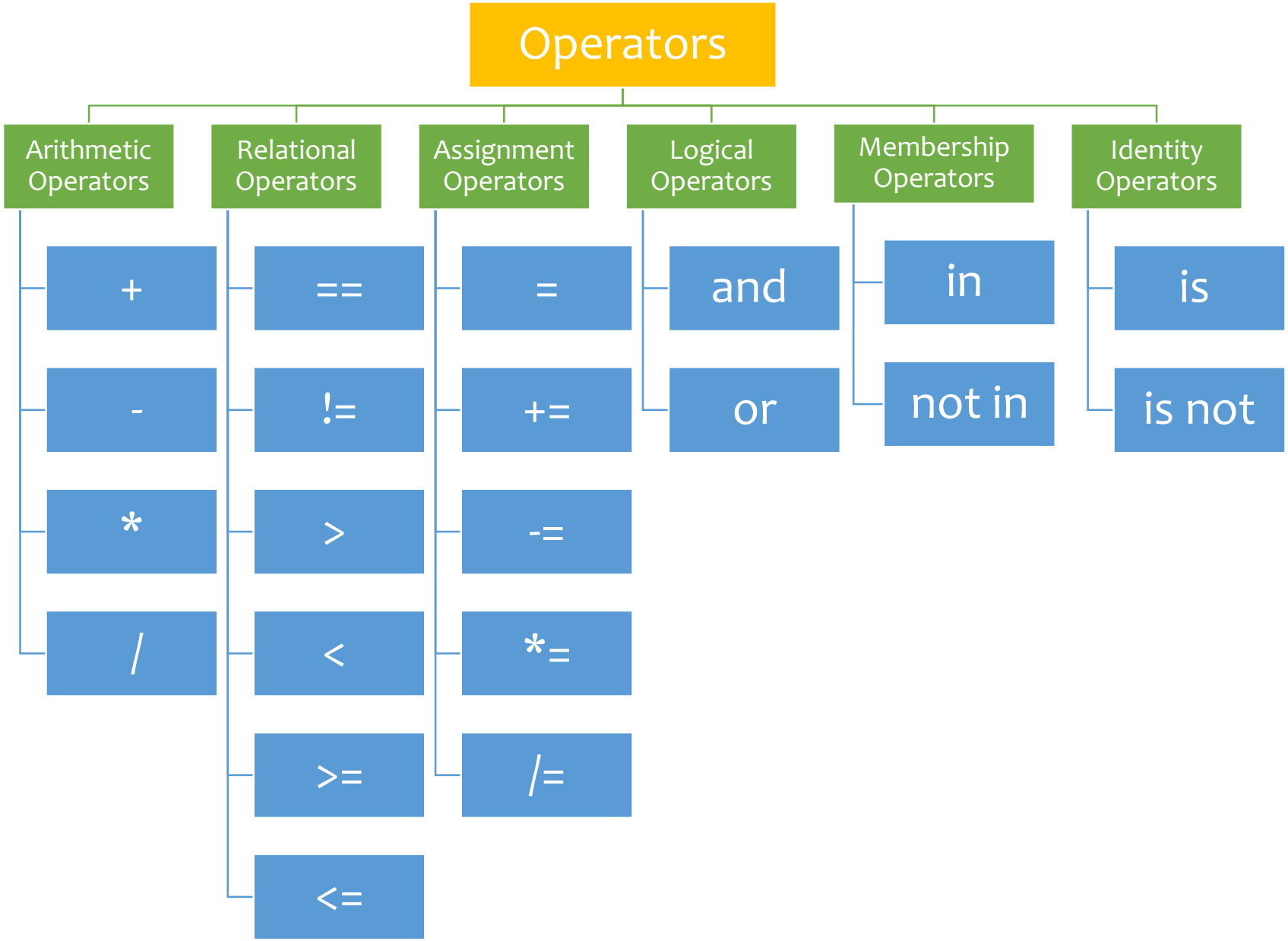
Data type

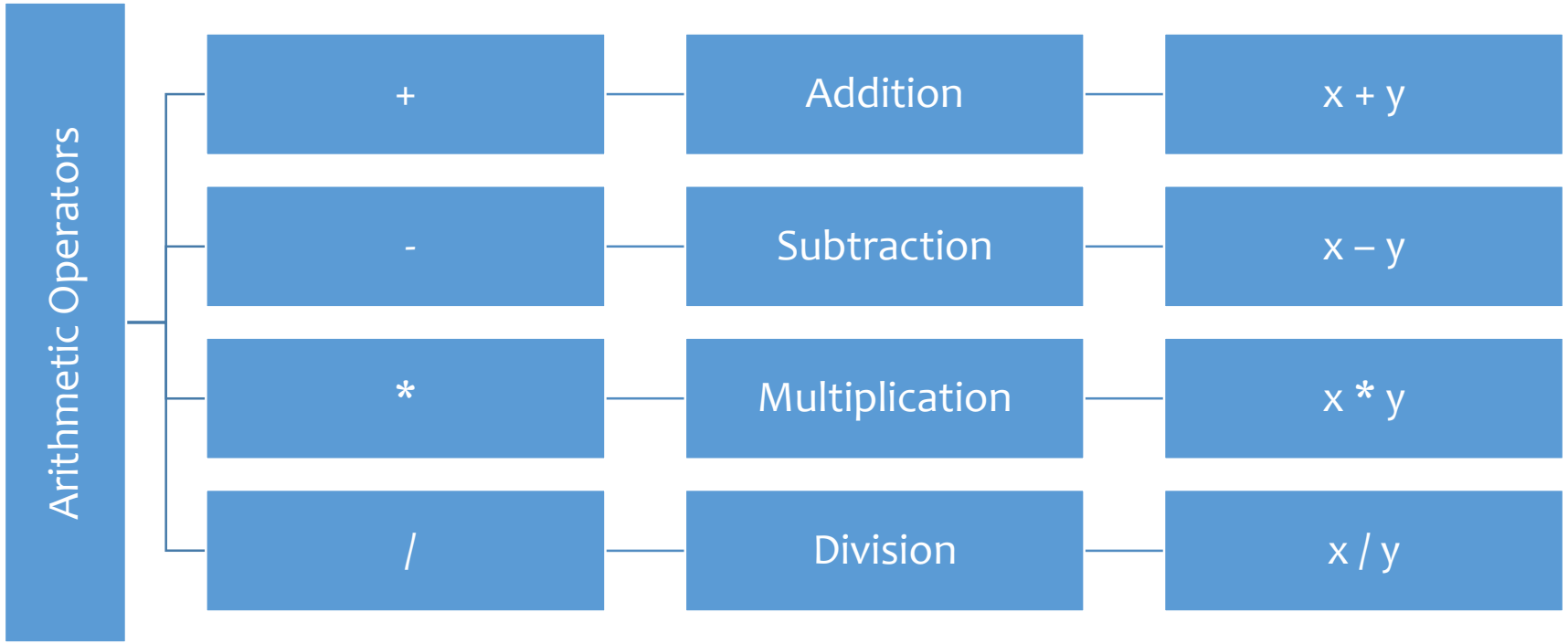
Python supports several **built-in data types,** each serving **different purposes.** Data types in Python are used to **represent different kinds** of data and **enable operations** specific to each type. Here are some common use cases for each data type:

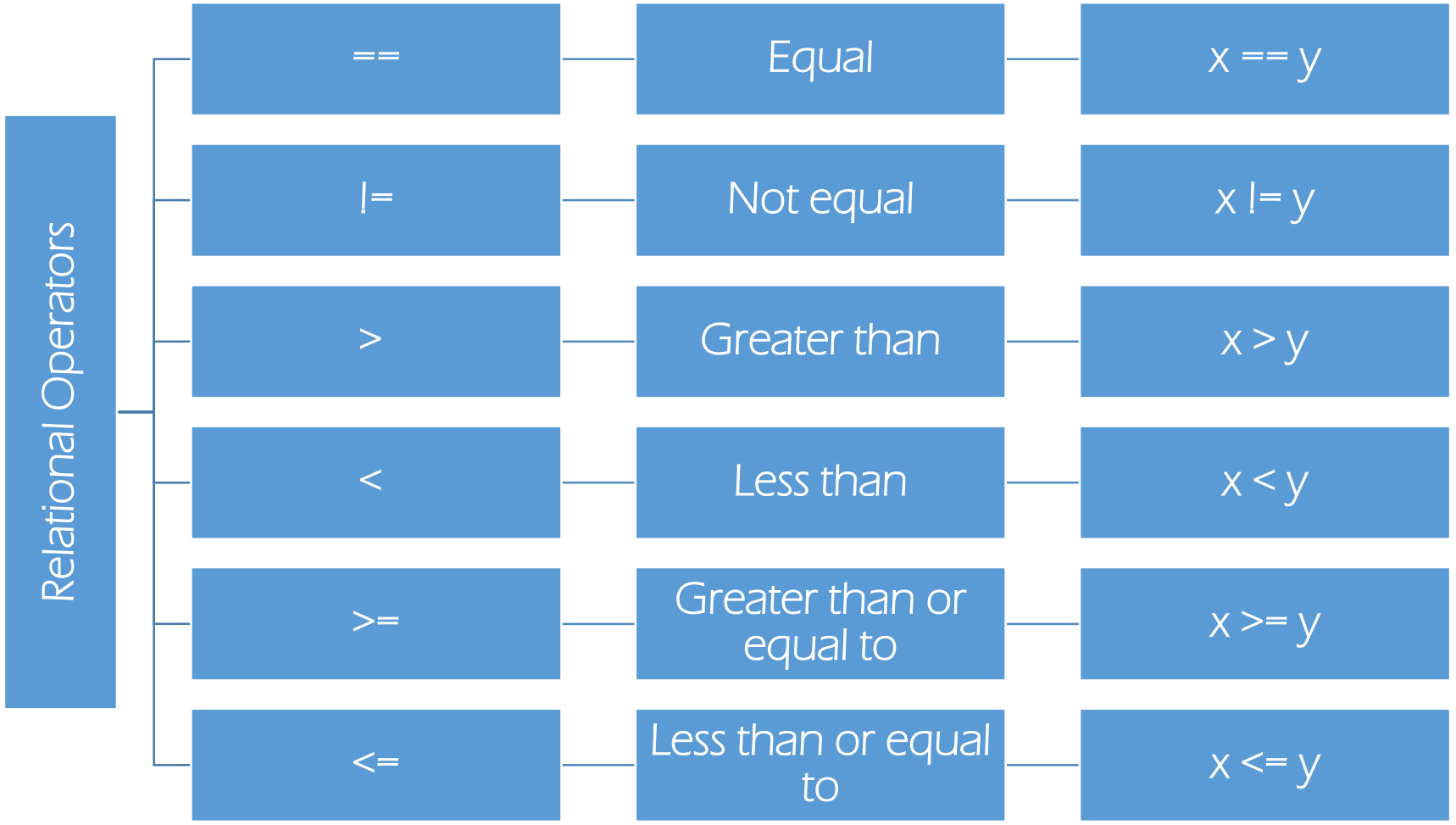
1) **Store Value**
2) **Classify the data**
3) **Perform operation**

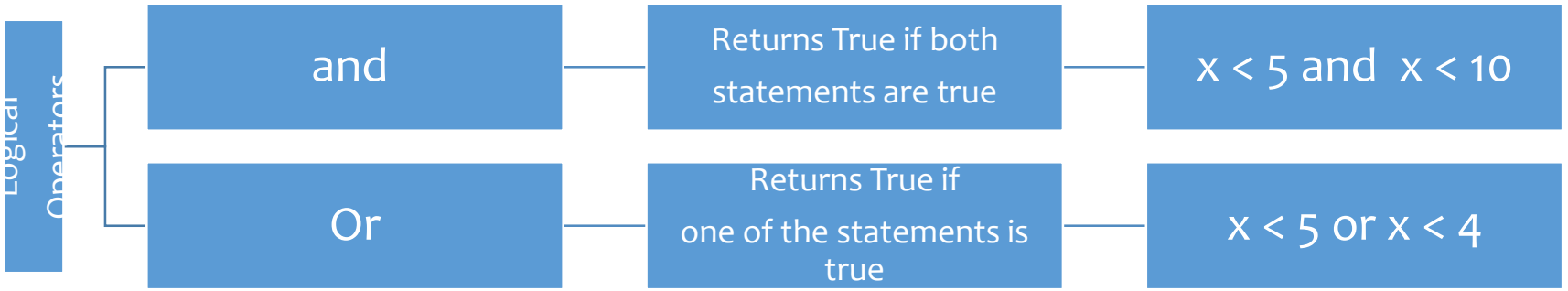

Type
Numeric
Int
X=5
Float
X = 5.5
Text
String
x = “Rajesh”
x = ‘Rajesh’
Bool
Bool
x = True
x = False
None
None
x = None

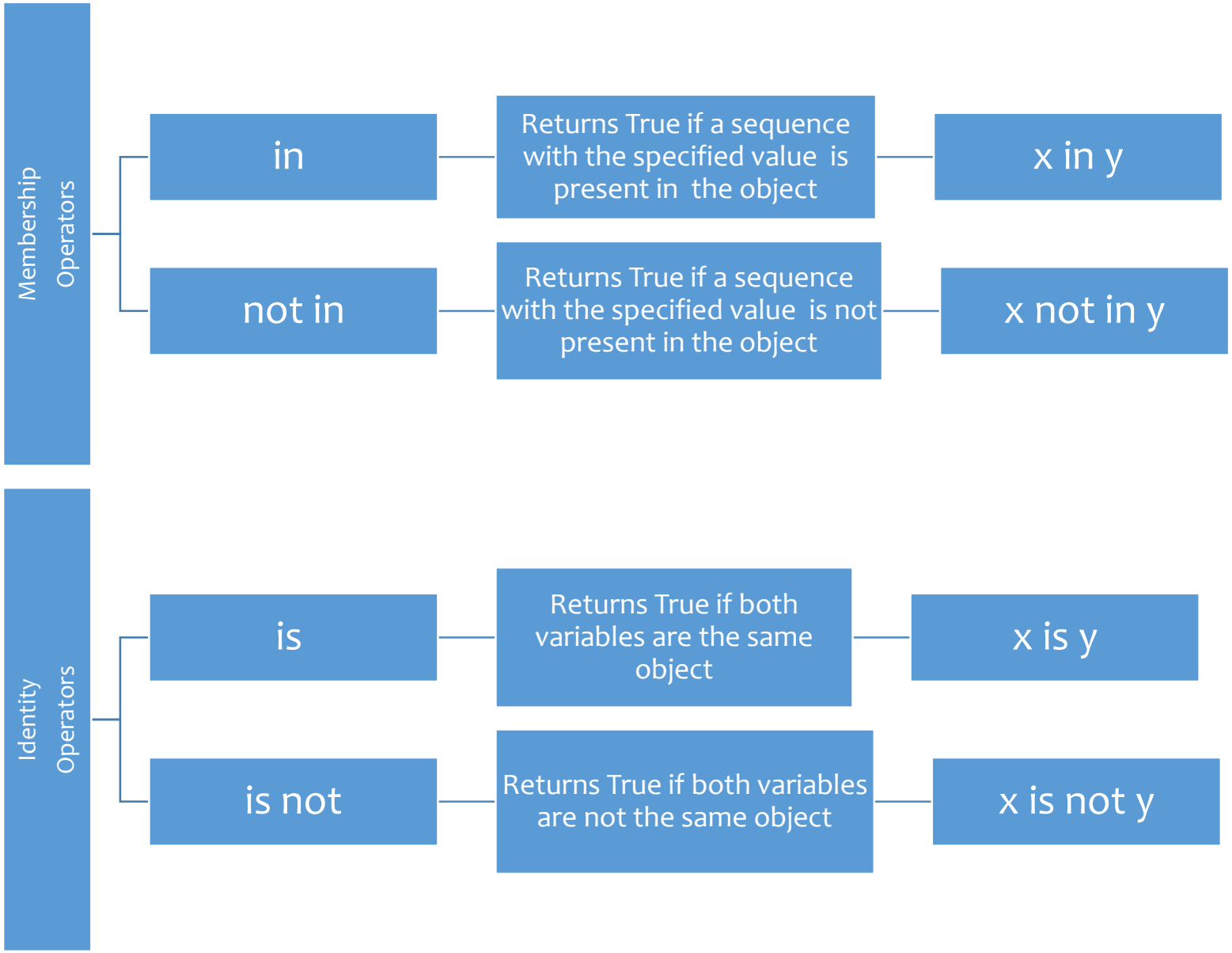

Type
Sequence
String
X = “Rajesh”
List
X = [1,2,3,4]
Tuple
X = (1,2,3,4)
Set Type
Set
X = {1,2,3,4}
Mapping Type
Dictionary
x = {'key1': 'value1', 'key2': 'value2'}


Operators
Arithmetic Operators
+
-
*
/
Relational Operators
==
!=
>
<
>=
<=
Assignment Operators
=
+=
-=
*=
/=
Logical Operators
and
or
Membership Operators
in
not in
Identity Operators
is
is not


Arithmetic Operators
+
Addition
x + y
-
Subtraction
x – y
*
Multiplication
x * y
/
Division
x / y


Relational Operators
==
Equal
x == y
!=
Not equal
x != y
>
Greater than
x > y
<
Less than
x < y
>=
Greater than or equal to
x >= y
<=
Less than or equal to
x <= y


Logical Operators
and
Returns True if both statements are true
x < 5 and x < 10
Or
Returns True if one of the statements is true
x < 5 or x < 4


Membership Operators
in
Returns True if a sequence with the specified value is present in the object
x in y
not in
Returns True if a sequence with the specified value is not present in the object
x not in y
Identity Operators
is
Returns True if both variables are the same object
x is y
is not
Returns True if both variables are not the same object
x is not y